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EXPERIMENT 6

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment

Implementation of VLAN and VLAN Trunking Protocols

Aim

To configure VLANs and VLAN Trunking in Cisco Packet Tracer, and to observe how VLANs logically partition a switched network while trunk links carry traffic for more than one VLAN between switches.

Objective

- To understand what a VLAN is and why it is used.
 - To distinguish between Layer 2 switching and Layer 3 switching.
 - To configure VLANs on Cisco switches.
 - To assign access ports to specific VLANs.
 - To understand the purpose of VLAN trunking.
 - To configure a trunk link between two switches.
 - To understand IEEE 802.1Q VLAN tagging.
 - To confirm that devices in the same VLAN can communicate.
 - To confirm that devices in different VLANs cannot communicate without Layer 3 routing.
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Software Requirements

Software

- Cisco Packet Tracer

Hardware (Virtual)

- Cisco 2960 Switches
- PCs
- Copper Straight-Through Cables

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Theory

A Layer 2 switch forwards Ethernet frames based on MAC addresses and operates at the Data Link Layer. A Layer 3 switch adds routing capability on top of that, so it can perform Inter-VLAN Routing in addition to normal Layer 2 switching.

A VLAN (Virtual Local Area Network) lets a single physical switch behave as several separate logical networks. For instance, VLAN 10 might be assigned to one department and VLAN 20 to another; devices in different VLANs are logically isolated from each other and need Layer 3 routing to communicate.

Using VLANs cuts down unnecessary broadcast traffic, keeps departments logically separated, makes the network easier to organize, and lets several logical networks share the same physical switching hardware.

An access port is normally tied to a single VLAN and is used to connect end devices such as PCs, printers, or servers.

A trunk port, by contrast, carries traffic for multiple VLANs over one physical link and is typically used between switches.

When VLAN traffic needs to cross a trunk, each frame is tagged with its VLAN information. The standard used for this is IEEE 802.1Q, which lets the receiving switch identify which VLAN a frame belongs to.

Practical / Experiment Steps

1. Open Cisco Packet Tracer.
 2. Place two Cisco 2960 switches and four PCs on the workspace.
 3. Cable the devices using Copper Straight-Through cables: PC1 → SW1 Fa0/1, PC2 → SW1 Fa0/2, PC3 → SW2 Fa0/1, PC4 → SW2 Fa0/2.
 4. Connect the two switches together: SW1 Gi0/1 → SW2 Gi0/1.
 5. Assign IP addresses to the PCs — PC1: 192.168.10.11/24, PC3: 192.168.10.12/24, PC2: 192.168.20.11/24, PC4: 192.168.20.12/24.
 6. Create VLAN 10 (HR) and VLAN 20 (SALES) on SW1.
 7. Assign SW1's Fa0/1 to VLAN 10 and Fa0/2 to VLAN 20.
 8. Repeat the VLAN creation and access-port assignment on SW2.
 9. Configure Gi0/1 on SW1 as a trunk port.
 10. Configure Gi0/1 on SW2 as a trunk port.
 11. Ping between PC1 and PC3 — both sit in VLAN 10, so the ping should succeed.
 12. Ping between PC2 and PC4 — both sit in VLAN 20, so the ping should succeed.
 13. Ping between PC1 and PC4 — they belong to different VLANs and no Layer 3 routing is configured, so the ping should fail.
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VLAN Configuration Commands

On each switch:

```
enable
configure terminal
```

```
vlan 10
name HR
exit

vlan 20
name SALES
exit
```

Access Port Configuration

For Fa0/1:

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit
```

For Fa0/2:

```
interface fa0/2
switchport mode access
switchport access vlan 20
exit
```

Trunk Configuration

On both switches:

```
interface gigabitEthernet 0/1
switchport mode trunk
exit
```

These commands together make up the complete SW1 and SW2 configuration used in this experiment.

Verification Commands

To view the configured VLANs:

```
show vlan brief
```

To confirm the trunk configuration:

```
show interfaces trunk
```

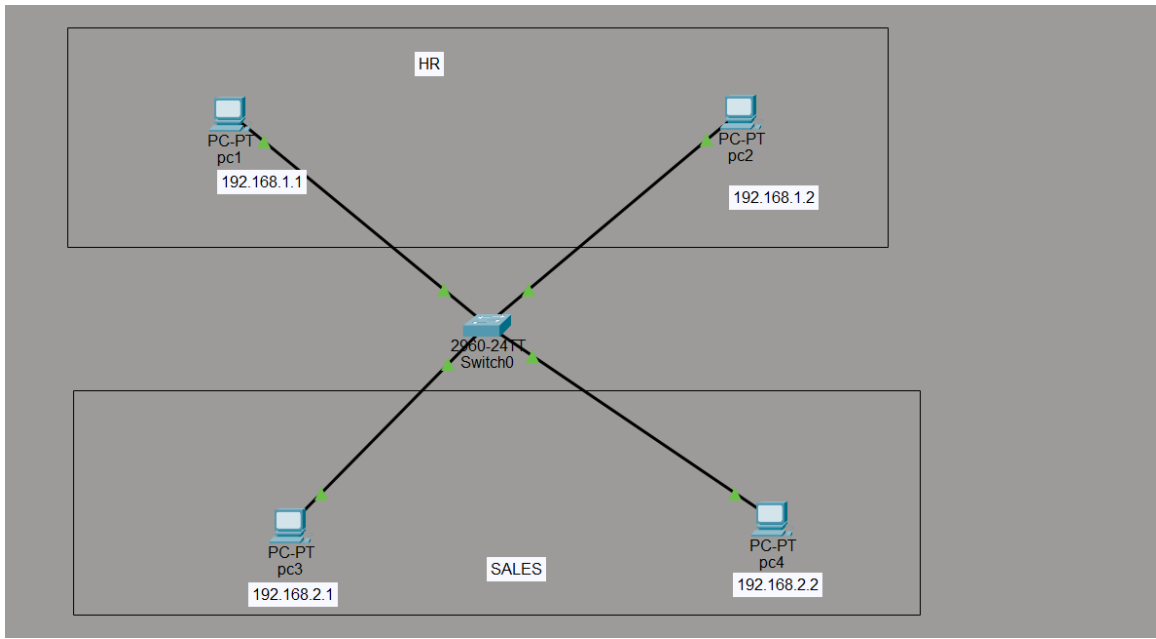
To inspect the MAC address table:

```
show mac address-table
```

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Output

Screenshot 1 – Network topology



Switch0 connects the HR group — pc1 (192.168.1.1) and pc2 (192.168.1.2) — and the SALES group — pc3 (192.168.2.1) and pc4 (192.168.2.2) — with each department's PCs mapped to its own VLAN off the same physical switch, so that HR and SALES traffic stays logically separated even though it rides the same hardware.

Learning Outcome

After completing this experiment, I was able to:

- Understand the concept and purpose of VLANs.
- Distinguish between Layer 2 and Layer 3 switches.
- Configure VLANs on Cisco switches.
- Assign access ports to specific VLANs.
- Understand the concept of VLAN trunking.
- Configure a trunk link between two switches.
- Understand the purpose of IEEE 802.1Q VLAN tagging.
- Verify VLAN and trunk configurations using Cisco IOS commands.
- Confirm communication between devices in the same VLAN.
- Understand why devices in different VLANs need Layer 3 routing to communicate.